



Subject Vision and Intent: Maths

Our Vision

At King's Cross Academy, we want every child to see themselves as a mathematician. We believe that Mathematics helps children make sense of patterns, relationships and problems within the world around them, and we want every child to develop a positive and confident relationship with the subject.

Our ambition is for children to leave KCA as confident, flexible and independent mathematical thinkers. We want them to be curious about mathematical ideas, make connections between what they know and approach unfamiliar problems with the confidence to explore possible solutions. Children learn that success in mathematics is not about always finding an answer immediately, but about thinking carefully, drawing upon what they know and persevering when something is challenging.

Our Mathematics curriculum reflects our Academy vision, *Love Learning Together*. Children explain their thinking, listen to the reasoning of others, explore different approaches and learn from mistakes. Mathematics also provides powerful opportunities to develop our KCA values: the courage to attempt something unfamiliar, the endurance to persevere, the integrity to recognise and learn from errors, and the aspiration to deepen understanding and achieve personal success.

Our Intent

Our Mathematics curriculum is ambitious, inclusive and carefully sequenced. It identifies the essential knowledge, vocabulary, concepts and mathematical relationships we want children to develop from the Early Years to Year 6, enabling them to build increasingly secure and connected mathematical understanding.

Learning is carefully broken down and sequenced so that new mathematical ideas build upon secure prior knowledge. Children are given regular opportunities to retrieve and apply what they have learned before, helping important knowledge become increasingly fluent and allowing them to make connections between different areas of mathematics.

Our curriculum develops conceptual understanding, fluency, mathematical reasoning and problem solving. These elements work together: secure knowledge and fluency give children

the foundations they need to reason, while reasoning and problem solving enable them to apply and deepen their mathematical understanding.

Children develop conceptual understanding through carefully chosen practical resources, representations and mathematical models. These help children see the structures and relationships within mathematics rather than simply learning procedures to follow. As understanding becomes secure, children become increasingly able to move flexibly between concrete resources, visual representations, mathematical language and abstract notation.

Fluency is developed through regular and purposeful practice. Children build secure knowledge of important number facts, relationships and procedures so that these can be recalled and applied accurately and efficiently. We want fluency to free children to think about increasingly complex mathematical ideas rather than place unnecessary demands on their working memory.

Mathematical reasoning is central to our curriculum. Children are expected to explain how they know, identify patterns and relationships, make generalisations, justify their thinking and consider whether an answer is reasonable. Through mathematical talk, children develop the confidence and precision to communicate their ideas and respond to the reasoning of others.

Problem solving provides opportunities for children to apply what they know in familiar and unfamiliar contexts. Children learn to identify relevant information, select appropriate strategies, test ideas and adapt their approach when necessary. We want children to develop the confidence to begin a problem even when the route to the solution is not immediately obvious.

Mathematical vocabulary is explicitly taught, modelled and revisited so that children develop the language needed to describe relationships, explain strategies and communicate their reasoning with increasing precision.

Every Child

We have high expectations for every child. Children with SEND, children who are disadvantaged and children who are developing English as an additional language access the same ambitious Mathematics curriculum.

Teachers identify the essential mathematical knowledge and concepts that all children need and adapt teaching so that barriers are reduced without lowering expectations. Explicit modelling, carefully chosen representations, manipulatives, structured mathematical talk, additional practice and opportunities to revisit prior learning enable children to develop secure understanding.

We recognise that children will sometimes require different amounts of time and support to secure an important mathematical idea. Where gaps in knowledge are identified, these are addressed precisely so that children can continue to access and connect with new learning rather than being unnecessarily restricted by what they have not yet mastered.

Children who grasp concepts rapidly are challenged to deepen their understanding through reasoning, connections, generalisation and increasingly complex problems rather than simply moving quickly through additional content.

We want every child to experience mathematical success. Mistakes are treated as valuable opportunities to identify misconceptions, reconsider thinking and develop understanding. Children learn that mathematical confidence grows through knowledge, practice, perseverance and the willingness to try again.

Mathematics Beyond the Lesson

Mathematical development at KCA extends beyond individual Mathematics lessons. Opportunities to retrieve, practise and apply important knowledge across the school week help children develop fluency and recognise the connections within mathematics.

Children encounter mathematics across the wider curriculum and in everyday contexts, applying their understanding of number, measurement, statistics, shape and spatial reasoning where it has a meaningful purpose. We want children to recognise that mathematics is not confined to a lesson but provides a powerful way of interpreting and understanding the world around them.

Through mathematical games, challenges and opportunities to celebrate achievement, we aim to develop enthusiasm for mathematics and give children further opportunities to experience the satisfaction of solving problems and becoming more fluent.

By the end of Year 6, our aim is not simply that children can perform mathematical procedures. We want them to understand the mathematics that underpins those procedures, recall and apply important knowledge fluently, reason about mathematical relationships and use what they know to solve unfamiliar problems.

Above all, we want children to leave KCA as confident, resilient and curious mathematicians who believe that mathematics is something they can understand, enjoy and succeed in, and who have the secure foundations they need for the next stage of their education and for mathematics in everyday life.